Light

GCSE Computer Science - Data Representation: Numbers (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the base of the binary number system?

- ○ **A.** Base 8
- ○ **B.** Base 10
- ○ **C.** Base 2
- ○ **D.** Base 16

Q2 (1 mark)

Why do computers use binary to store and process data?

- ○ **A.** Binary numbers are smaller than denary numbers

- ○ **B.** Binary was invented specifically for modern computers
- ○ **C.** Programmers find binary easier to read than denary
- ○ **D.** Electronic circuits can easily represent two states: on (1) and off (0)

Q3 (1 mark)

What is the denary (decimal) value of the binary number 1010?

- ○ **A.** 8
- ○ **B.** 12
- ○ **C.** 10
- ○ **D.** 20

Q4 (1 mark)

What is the binary representation of the denary number 7?

- ○ **A.** 101
- ○ **B.** 110
- ○ **C.** 1000
- ○ **D.** 111

Q5 (1 mark)

Hexadecimal is a base 16 number system. Which symbols does it use?

- ○ **A.** 0-9 and A-F
- ○ **B.** 0-9 only
- ○ **C.** A-P
- ○ **D.** 0-15

Q6 (1 mark)

In hexadecimal, what denary value does the letter C represent?

- ○ **A.** 3
- ○ **B.** 10
- ○ **C.** 13
- ○ **D.** 12

Q7 (1 mark)

How many bits are in a nibble?

- ○ **A.** 4
- ○ **B.** 8
- ○ **C.** 2
- ○ **D.** 16

Q8 (1 mark)

In the binary number 10110100, which bit is the most significant bit (MSB)?

- ○ **A.** The rightmost 0
- ○ **B.** The middle bit
- ○ **C.** Any of the 1s
- ○ **D.** The leftmost 1

Q9 (1 mark)

In the binary number 11010011, which bit is the least significant bit (LSB)?

- ○ **A.** The rightmost 1
- ○ **B.** The second bit from the right
- ○ **C.** The first 0 from the left
- ○ **D.** The leftmost 1

Q10 (1 mark)

What is the maximum denary value that can be stored in 8 bits?

- ○ **A.** 128
- ○ **B.** 255
- ○ **C.** 256

- ○ **D.** 200

Q11 (1 mark)

Why is hexadecimal often used instead of binary when working with computers?

- ○ **A.** Hexadecimal is more compact and easier for humans to read than long binary strings
- ○ **B.** Hexadecimal is the only number system computers understand
- ○ **C.** Computers process hexadecimal faster than binary
- ○ **D.** Binary cannot represent large numbers

Q12 (1 mark)

What is the result of the binary addition 0101 + 0010?

- ○ **A.** 0110
- ○ **B.** 1010
- ○ **C.** 1000
- ○ **D.** 0111

Q13 (2 marks)

Convert the denary number 156 to an 8-bit binary number.

- A. 10011100
- B. 10011000
- C. 10101100
- D. 11001100

Q14 (2 marks)

Convert the binary number 11100101 to denary.

- A. 213
- B. 225
- C. 229
- D. 245

Q15 (2 marks)

Convert the hexadecimal number 3F to denary.

- A. 63
- B. 47
- C. 53
- D. 79

Q16 (2 marks)

Convert the denary number 200 to hexadecimal.

- ○ **A.** C0
- ○ **B.** D0
- ○ **C.** C8
- ○ **D.** D8

Q17 (2 marks)

Convert the binary number 10110010 to hexadecimal.

- ○ **A.** A2
- ○ **B.** B2
- ○ **C.** C2
- ○ **D.** D2

Q18 (2 marks)

Convert the hexadecimal number 5D to binary.

- ○ **A.** 01011011
- ○ **B.** 01011111
- ○ **C.** 01011101

- D. 01110101

Q19 (2 marks)

What is the result of the binary addition 01101001 + 00110101?

- A. 10011100
- B. 10011110
- C. 10100100
- D. 10111110

Q20 (2 marks)

What causes an overflow error in binary addition?

- A. Adding two odd numbers together
- B. Adding a number to itself
- C. When the result requires more bits than are available to store it
- D. When the result is exactly 256

Q21 (2 marks)

What is the effect of a left binary shift by one place on the number 00010100?

- A. The value is halved to 00001010

- B. The value becomes 10100000
- C. The value stays the same
- D. The value is doubled to 00101000

Q22 (2 marks)

What is the effect of a right binary shift by two places on the number 01100000?

- A. The value is divided by 4 to 00011000
- B. The value is multiplied by 4 to 10000000
- C. The value is halved to 00110000
- D. The value is divided by 2 to 00110000

Q23 (2 marks)

How many bytes are in 4 kilobytes (KB)?

- A. 400
- B. 1,024
- C. 4,000
- D. 4,096

Q24 (3 marks)

Calculate 11010110 + 01011010 in binary.

Explain whether an overflow error occurs if using an 8-bit register.

- ○ **A.** Result is 00110000; no overflow occurs
- ○ **B.** Result is 100110000; overflow occurs because the result needs 9 bits but only 8 are available
- ○ **C.** Result is 11111111; no overflow occurs
- ○ **D.** Result is 10110000; overflow occurs because both numbers are negative

Q25 (3 marks)

A programmer wants to multiply a number by 10 using only binary shifts and addition.
Analyse how this can be achieved.

- ○ **A.** It is impossible to multiply by 10 using shifts
- ○ **B.** Shift left 3 times to multiply by 8, then shift left 1 time to multiply by 2, giving 16
- ○ **C.** Shift left 3 times to get 8x the original, then add the original shifted left once (2x), giving 8x + 2x = 10x
- ○ **D.** Shift left 10 times to multiply by 10

Q26 (3 marks)

Web colours are often specified in hexadecimal as #RRGGBB where RR, GG, BB represent red, green, blue values.
Analyse what colour the code #FF8000 represents and explain why hex is used.

- ○ **A.** White; hex is used because websites require hexadecimal

○ **B.** Orange (full red, half green, no blue); hex is used because it compactly represents the 0-255 range for each colour channel using just two characters

○ **C.** Purple (full red, no green, full blue); hex is faster to process than binary

○ **D.** Black; hex is used because monitors only understand hexadecimal

Q27 (3 marks)

A student writes the binary number 11010 and another writes 00011010. Analyse whether these represent the same value and explain the significance of leading zeros.

○ **A.** They are different values; leading zeros change the number's meaning

○ **B.** They are the same only when converted to denary

○ **C.** They are different because one is 5-bit and one is 8-bit binary

○ **D.** They are the same value (26); leading zeros do not change value but ensure consistent bit-width for storage and processing

Q28 (3 marks)

Convert the denary number 187 to both binary and hexadecimal. Verify your hexadecimal answer by converting it back to denary.

○ **A.** Binary: 10111011, Hex: B7; Verification: (11 x 16) + 7 = 176 + 7 = 183

○ **B.** Binary: 10111011, Hex: BB; Verification: (11 x 16) + 11 = 176 + 11 = 187

○ **C.** Binary: 10110111, Hex: B7; Verification: (11 x 16) + 7 = 176 + 7 = 183

○ **D.** Binary: 10111111, Hex: BF; Verification: (11 x 16) + 15 = 176 + 15 = 191

Q29 (3 marks)

The binary number 10110101 is shifted right by 3 places.
Analyse what happens to the value and identify any data loss.

○ **A.** Result: 00010110, value approximately divided by 8, three bits (101) are lost causing the result to be truncated from 22.625 to 22

○ **B.** Result: 00010110, value divided by 8, no data loss occurs

○ **C.** Result: 10100000, value multiplied by 8, no data loss

○ **D.** Result: 00010110, value unchanged, bits simply moved

Q30 (3 marks)

A game stores player scores using unsigned binary numbers.
Analyse the storage requirements and maximum score possible with different bit allocations: 8-bit, 16-bit, and 24-bit.

○ **A.** 8-bit: max 128; 16-bit: max 32,768; 24-bit: max 8,388,608

○ **B.** 8-bit: max 255; 16-bit: max 65,535; 24-bit: max 16,777,215

○ **C.** 8-bit: max 256; 16-bit: max 65,536; 24-bit: max 16,777,216

○ **D.** 8-bit: max 255; 16-bit: max 65,000; 24-bit: max 16,000,000

Total: **55** marks • Time: **30 min**

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